



Chair of
Information Theory
and Data Analytics

RWTHAACHEN
UNIVERSITY

RESOURCE-EFFICIENT INFERENCE IN LARGE GENERATIVE MODELS

PATRICK TAZO KUETE

CHAIR OF INFORMATION THEORY AND DATA ANALYTICS
RWTH AACHEN UNIVERSITY



Chair of
Information Theory
and Data Analytics

RWTHAACHEN
UNIVERSITY

RESOURCE-EFFICIENT INFERENCE IN LARGE GENERATIVE MODELS

PATRICK TAZO KUETE

A seminar submitted to the
Faculty of Electrical Engineering and Information Technology,
RWTH Aachen University,

reviewed by Univ.-Prof. Dr. Ing. Anke Schmeink and supervised by Hazem el Hadj Khalifa,
M.Sc.

CHAIR OF INFORMATION THEORY AND DATA ANALYTICS
RWTH AACHEN UNIVERSITY

Eidesstattliche Versicherung

Name, Vorname

Matrikelnummer

Ich versichere hiermit an Eides Statt, dass ich die vorliegende Seminararbeit mit dem Titel

Resource-Efficient Inference in Large Generative Models

selbständig und ohne unzulässige fremde Hilfe erbracht habe. Ich habe keine anderen als die angegebenen Quellen und Hilfsmittel benutzt. Für den Fall, dass die Arbeit zusätzlich auf einem Datenträger eingereicht wird, erkläre ich, dass die schriftliche und die elektronische Form vollständig übereinstimmen. Die Arbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen.

Aachen, den 18th June 2026

Ort, Datum

Unterschrift

Belehrung:

§ 156 StGB: Falsche Versicherung an Eides Statt

Wer vor einer zur Abnahme einer Versicherung an Eides Statt zuständigen Behörde eine solche Versicherung falsch abgibt oder unter Berufung auf eine solche Versicherung falsch aussagt, wird mit Freiheitsstrafe bis zu drei Jahren oder mit Geldstrafe bestraft.

§ 161 StGB: Fahrlässiger Falscheid; fahrlässige falsche Versicherung an Eides Statt

(1) Wenn eine der in den §§ 154 bis 156 bezeichneten Handlungen aus Fahrlässigkeit begangen worden ist, so tritt Freiheitsstrafe bis zu einem Jahr oder Geldstrafe ein.

(2) Strafflosigkeit tritt ein, wenn der Täter die falsche Angabe rechtzeitig berichtigt. Die Vorschriften des § 158 Abs. 2 und 3 gelten entsprechend

Die vorstehende Belehrung habe ich zur Kenntnis genommen:

Aachen, den 18th June 2026

Ort, Datum

Unterschrift

Abstract

A short summary.

Acknowledgements

Thanks to the world!

Contents

Acronyms	xiii
Notation glossary	xv
1 Introduction	1
1.1 Motivation	1
1.2 Aims	1
1.3 Thesis structure	2
2 Theoretical foundations	3
2.1 Mathematical optimization	3
2.1.1 Optimization problem in standard form	3
2.1.2 Non-negative least squares	3
Bibliography	5

Acronyms

NNLS non-negative least squares 3

Notation glossary

$n \in \mathbb{N}$	Natural number
$a \in \mathbb{R}$	Real number, scalar
$a \in \mathbb{R}_{\geq 0}$	Non-negative real number, non-negative scalar
$\mathbf{a} \in \mathbb{R}^n$	Real vector of size n
$\mathbf{a} \in \mathbb{R}^{n \times m}$	Real matrix of size $n \times m$
$\mathfrak{P}(\mathbb{R}^n)$	Set with all subsets of $\mathbb{R}^n$ as elements
$:=$	Is defined as
$a_i, [\mathbf{a}]_i$	i th element in vector $\mathbf{a}$
$a_{ij}, [\mathbf{a}]_{ij}$	Element at i th row and j th column in matrix $\mathbf{a}$
$\mathbf{a}_{i:}, [\mathbf{a}]_{i:}$	i th row in matrix $\mathbf{a}$
$\mathbf{a}_{:,j}, [\mathbf{a}]_{:,j}$	j th column in matrix $\mathbf{a}$
$\mathbf{b} := \text{vec}(\mathbf{a}) \in \mathbb{R}^{nm}$	Vector $\mathbf{b}$ contains all elements of $\mathbf{a}$ in row-wise order
$\emptyset$	Empty set
$\mathcal{I} \subset \mathbb{N}$	Set of indices
$ \mathcal{I} $	Cardinality of set $\mathcal{I}$
$\mathbf{b} := [\mathbf{a}]_{\mathcal{I}} \in \mathbb{R}^{ \mathcal{I} }$	Vector containing elements $\mathcal{I}$ of vector $\mathbf{a}$ with indices in ascending order
$[\mathbf{a}]_{i\mathcal{I}} = [\mathbf{a}_{i:}]_{\mathcal{I}}, [\mathbf{a}]_{\mathcal{I}j} = [\mathbf{a}_{:,j}]_{\mathcal{I}}$	Vector containing elements $\mathcal{I}$ of i th row or j th column respectively
$\mathbf{b} := [\mathbf{a}]_{\mathcal{I}_1 \mathcal{I}_2} \in \mathbb{R}^{ \mathcal{I}_1 \times \mathcal{I}_2 }$	Matrix containing elements at all combinations of given row and column indices $\mathcal{I}_1$ and $\mathcal{I}_2$ respectively
$\mathbf{e}_i$	Unit vector i
$\mathbf{i}_n$	Identity matrix of size $n \times n$
$\mathbf{0}_n, \mathbf{0}_{n \times m}$	Zero-vector of size n or zero-matrix of size $n \times m$

$\mathbf{1}_n, \mathbf{1}_{n \times m}$	One-vector of size n or one-matrix of size $n \times m$
$\mathbf{A}^\top$	Transposed matrix
$\mathbf{A}^{-1}$	Matrix inverse
$\mathbf{A}^\dagger$	Moore-Penrose pseudo-inverse
$\text{diag}(\mathbf{a})$	Diagonal matrix with elements of $\mathbf{a}$
$\text{diag}(\mathbf{a})$	Diagonal matrix with diagonal elements of $\mathbf{a}$
$ \cdot $	Absolute value
$\ \cdot\ _2$	Euclidean norm
$\ \cdot\ _F$	Frobenius norm
$\ \cdot\ _0$	l_0 -norm, i.e., number of non-zero elements
$[x]^+ = \max(x, 0)$	Positive part of x
$[x]^- = \min(x, 0)$	Negative part of x
$\neg a$	Logical NOT
$a \wedge b$	Logical AND
$a \vee b$	Logical OR
$a \oplus b$	Logical XOR

1 Introduction

In this chapter the thesis is introduced. Section 1.1 presents the motivation for the chosen topic, section 1.2 describes the aims of the thesis and section 1.3 outlines the content.

1.1 Motivation

Since the introduction of deep learning, the field of machine learning has seen a tremendous growth. In particular, large generative models have shown impressive capabilities in various domains, such as natural language processing and computer vision. However, these models often require significant computational resources for inference, which limits their accessibility and applicability in real-world scenarios. Therefore, there is a growing interest in developing resource-efficient inference techniques for large generative models. This thesis aims to explore and evaluate various approaches to achieve resource-efficient inference while maintaining the performance of large generative models.

1.2 Aims

The main aims of this thesis are as follows:

- To review and analyze existing techniques for resource-efficient inference in large generative models.
- To propose novel methods for improving the efficiency of inference in large generative models.
- To evaluate the proposed methods on benchmark datasets and compare their performance with existing techniques.
- To provide insights and recommendations for future research in the field of resource-efficient inference in large generative models.

1.3 Thesis structure

The thesis is structured as follows: ?? provides a theoretical background on large generative models and resource-efficient inference techniques. ?? describes the proposed methods for improving inference efficiency. Finally, ?? summarizes the findings and discusses future research directions.

2 Theoretical foundations

2.1 Mathematical optimization

First a general optimization problem in standard form is formulated (section 2.1.1). Afterwards, a non-negative least squares (NNLS) problem is presented. This section is based on [1].

2.1.1 Optimization problem in standard form

The purpose of mathematical optimization is to find an $\mathbf{x}$ that minimizes $f_0(\mathbf{x})$ under some constraints. The vector $\mathbf{x} \in \mathbb{R}^n$ is called *optimization variable* and $f_0 : \mathbb{R}^n \rightarrow \mathbb{R}$ *objective function*. The constraints split into *inequality constraints* $f_i(\mathbf{x}) \leq 0$ and *equality constraints* $h_i(\mathbf{x}) = 0$ with $f_i, h_i : \mathbb{R}^n \rightarrow \mathbb{R}$. This leads to the optimization problem in *standard form*

$$\begin{aligned} & \underset{\mathbf{x}}{\text{minimize}} && f_0(\mathbf{x}) \\ & \text{subject to} && f_i(\mathbf{x}) \leq 0, \quad i = 1 \dots m \\ & && h_i(\mathbf{x}) = 0, \quad i = 1 \dots p. \end{aligned} \tag{2.1.1}$$

A point $\mathbf{x}$ is called *feasible*, if it satisfies all constraints. The *optimal value* p^* is defined as

$$p^* = \inf \left\{ f_0(\mathbf{x}) \mid f_i(\mathbf{x}) \leq 0, \quad i = 1 \dots m, \quad h_i(\mathbf{x}) = 0, \quad i = 1 \dots p \right\}.$$

2.1.2 Non-negative least squares (NNLS)

The NNLS problem is a special optimization problem with several applications in different fields. It denotes as

$$\begin{aligned} & \underset{\mathbf{x}}{\text{minimize}} && \frac{1}{2} \|\mathbf{a}\mathbf{x} - \mathbf{b}\|_2^2 \\ & \text{subject to} && \mathbf{x} \geq \mathbf{0}_n. \end{aligned} \tag{2.1.2}$$

Comparing (2.1.2) with the optimization problem in standard form (2.1.1), the objective function $f_0 : \mathbb{R}^n \rightarrow \mathbb{R}$ and inequality constraint functions $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$, $i = 1 \dots n$ are

$$\begin{aligned} f_0(\mathbf{x}) &= \frac{1}{2} \|\mathbf{a}\mathbf{x} - \mathbf{b}\|_2^2 \\ f_i(\mathbf{x}) &= -x_i, \quad i = 1 \dots n \end{aligned}$$

Bibliography

- [1] S. Boyd and L. Vandenberghe, “Convex optimization,” 2004.